CHAPTER ONE

COMPUTER HARDWARE

Concepts : Hardware

- **The term hardware refers to the physical components of your computer such as the system unit, mouse, keyboard, monitor etc.**

What is a personal computer?

- IBM invented the PC (Personal Computer) way back in 1981. All PCs released since then are in many ways compatible with the original design, though many extensions have been made. The term PC compatible relates to PCs manufactured by companies other than IBM which are compatible with the traditional PC specification. In the early days, most PCs ran an operating system called DOS (Disk Operating System). These days most PCs will be running a version of Microsoft Windows such as Microsoft Vista.

Capacity: Large hard disks combined with a large working memory (RAM)

Speed: Fast. Normally measured in GHz.

Costs: Getting cheaper by the day.

Laptop & palmtop computers

- Laptop computers, as the name implies, are small portable computers which can run on batteries as well as mains power. They use special screens, rather than the traditional bulky VDUs (Visual Display Units), which allows for longer battery life as well as portability. A newer term, "Notebooks", simply indicates a VERY small laptop. These are especially popular with salespersons on the move or people giving presentations. While they tend to still be more expensive than an equivalent Desktop computer, they can now match the power of a Desktop computer. Palmtops are even smaller computers which can literally fit into the palm of your hand.

Capacity: Large hard disks combined with a large working memory (RAM) -Often less powerful than for a PC of equivalent price.

Speed: Fast. Normally measured in GHz. Often speed specifications are less than for a PC of equivalent price.

Costs: Components need to be much more compact, so there is a price overhead when compared to a PC of equivalent power.

Typical Users: Business users, people on the move, educational users.

Features of handheld portable digital devices

- A range of small portable digital devices are now available including:
  - Personal digital assistants (PDAs)
  - Mobile (Cell) phones
  - Smartphone's
  - Multimedia players

Personal Digital Assistant (PDA)

- These devices use a special pen, rather than a keyboard and can be used for storing and retrieving information. Like most computer devices, many can connect to the Internet. They are extremely compact.
- PDAs have many built-in features such as a calculator, a clock and a calendar.
- Many allow you to send and receive emails and even browse the web.
- PDAs can also be used for accessing the Internet, sending and receiving emails, video recording, typewriting and word processing, use as an address book, making and writing on spreadsheets, scanning bar codes, playing computer games, recording survey responses, and Global Positioning System (GPS). Newer PDAs also have both color screens and audio capabilities, enabling them to be used as mobile phones (smart phones), web browsers, or portable media players. Many PDAs can access the Internet, intranets or extranets via Wi-Fi, or

Parts of a computer

- You should understand some of the basic elements that make up a computer including:
  - Central processing unit (CPU)
  - Types of memory
  - The hard disk
  - Input and output devices

The CPU

- The CPU (Central Processing Unit) is normally an Intel (or equivalent) and it is one of the most important components within your computer. It determines how fast your computer will run and the CPU speed is measured by its MHz or GHz speed. Thus, a 2 GHz Pentium is much faster than say a 1 GHz Pentium CPU. It is the CPU which performs all the calculations within the computer, when running programs such as word-processors,

spreadsheets and databases. The CPU contains following elements:

Control Unit:

The control unit is responsible for controlling the sequencing and timing of the other elements making up the CPU.

Arithmetic Logic Unit (ALU):

The ALU performs the mathematical calculations using data stored within the CPU registers.

Registers:

The registers are memory storage areas within the CPU that hold the data that is worked on by the ALU.

BUS:

The computer bus transports data between the memory and registers.

More information:

Intel: http://www.intel.com AMD: http://www.amd.com

Memory (RAM)

- The RAM (Random Access Memory) within your computer is where the operating system is loaded to when you switch on your computer and also where your applications are copied to when you start an application, such as a word processor or database program. When you create data, (e.g. letters and pictures), these

are initially created and held in RAM and then copied to disk when you save the data. As a rule of thumb, the more RAM you have installed in your computer the better. These days you will commonly find over 512 Megabytes of RAM installed.

ROM-BIOS

- The ROM-BIOS (Read Only Memory - Basic Input Output System) chip is a special chip held on your computer's system (mother) board. It contains software which is required to make your computer work with your operating system, for instance it is responsible for copying your operating system into RAM when you switch on your computer.

Disks

- You use disks to store any data that you create. This can range from a memo created within a word processor to a video file created using a video camera. There are many different types of disk.

The Hard (Fixed) Disk

- Hard disks are the main, large data storage areas within your computer. Hard disks are used to store your operating system, your application programs (i.e. your word processor, games etc) and your data. They are much faster than CD/DVDs and can also

hold much more data. Hard disks are installed within the system unit of your computer.

Input/output ports

- The Input and Output ports are normally located at the back or on the side of your computer. These include ports such as USB, serial, parallel, network and FireWire ports. You plug cables into these ports to connect your computer to other devices, such as printers, scanners and cameras.

Universal Serial Bus (USB) Port

- The Universal Serial Bus is a relatively new item within the PC. You will see one or more USB sockets at the back of the system unit, allowing you to plug in devices designed for the USB. These devices include printers, scanners and digital cameras. Memory sticks can also be plugged into a USB port allowing you to copy data to or from your hard disk.

Serial Port

- The serial port is a socket located at the back of your computer which enables you to connect items to the computer, such as a modem. They are commonly labelled as COM1 or COM2.

Network Port

- The network port allows you to plug a 'network cable' into your computer, which then lets you communicate with other computers connected to your local network or to other computers via the Internet.

FireWire Port

- FireWire is an interface from Apple Inc. that allows high data transfer between your computer and a compatible device such as a digital camera. FireWire has largely replaced earlier ways of transferring data, such as parallel or SCSI cables. There are numerous versions of FireWire including fibre optic, coaxial and wireless versions. Most multimedia computers will have FireWire ports built into them. Firewire connectors usually look like this:

Computer Performance

Factors affecting computer performance

- There are a wide range of factors that can affect the performance of your computer. These include CPU speed, RAM size, type of graphics card processor and memory plus the number of applications running. It is important to realise that it is not just the speed of the CPU that affect the overall performance of your computer. There is no point in having a very fast CPU if the other parts of a computer may slow down the real world performance.

CPU Clock speed

- The computer clock speed governs how fast the CPU will run. The higher the clock speed the faster the computer will work for you. The clock speed is given in megahertz (MHz). The higher the MHz speed the faster the computer.

RAM size

- As a rule the more memory you have the faster the PC will appear to operate. Windows also uses the hard disk a lot, so logically the faster the hard disk can operate then again the faster the PC will appear to run.

Hard disk speed and storage

- Hard disks are also measured by their speed, defined by the disk access time, which is measured in milliseconds. The smaller this access time the faster the hard disk will store or retrieve data. The data storage capacity of hard disks continues to increase as new products are released. The disk storage capacity is measured in Gigabytes (GBytes). 1 GByte is equivalent to 1024 Mbytes.

Free hard disk space

- To get the most out of your Windows based PC, you not only need a fast hard disk but also a large hard disk with plenty of "spare space". This is due to the fact Windows is constantly moving data between the hard disk and RAM (Random Access Memory). Microsoft Windows will create many so-called "temporary files" which it uses for managing your programs. In fact, if you have very little free hard disk space you may find that Microsoft Windows will not be able to run your programs at all.

De-fragmenting files

- If you are running Windows you may find that if you click on the **Start** menu, select **Programs,** and then select the **Accessories / System tools** group, there is a de-fragmentation program. Running this periodically may noticeably speed up the operation of your PC. When you use a PC, over a period of time the files

CPU speeds

- The speed (operating frequency) of the CPU is measured in megahertz (MHz) or gigahertz (GHz). The higher the value the faster the CPU will operate. The original IBM PC released way back in 1981 ran at 4.77 MHz whereas modern PCs can run at over 4000 MHz, which gives you an idea of how far things have progressed.
- 1 MHz means that the device will run at one million cycles per second. 1 GHz is a thousand times faster, so that 1 GHz = 1000 MHz.

Memory and Storage

What is computer memory?

- You can store data on your hard disk, while data which is being processed is stored in RAM (Random Access Memory). Data which is stored on a hard disk can be permanent, while data in RAM is only temporary. Normally when people talk about memory in relation to a PC, they are talking about RAM.

RAM

- Random Access Memory (RAM) is the main 'working' memory used by the computer. When the operating system loads from disk when you first switch on the computer, it is copied into RAM. The original IBM PC could only use up to 640 KB of memory (just over half a megabyte), whereas a modern computer can effectively house as much RAM as you can afford to buy. Commonly modern computers are supplied with over 128 MB of RAM. As a rough rule, a Microsoft Windows based computer will operate faster if you install more RAM. When adverts refer to a computer having 128 Mbytes of memory, it is this RAM which they are talking about. Data and programs stored in RAM are volatile (i.e. the information is lost when you switch off the computer).

ROM

- Read Only Memory (ROM) as the name suggests is a special type of memory chip which holds software which can be read but not written to. A good example is the ROM-BIOS chip, which contains read-only software. Often network cards and video cards also contain ROM chips.

ROM-BIOS

- The 'Read Only Memory Basic Input Output System' chip is a chip located on the computer's system (mother) board, which contains software. This software performs a variety of tasks. When you first switch on the computer the ROM-BIOS software performs a self-diagnostic to check that the computer is working

OK. This software then loads your operating system from the disk into the RAM.

Video (graphics) memory

- The picture which you see on your screen is a form of data and this data has to be stored somewhere. The on-screen pictures are held in special memory chips called video memory chips; these chips are usually located on the video card. A modern computer will be supplied with several Megabytes of video memory.

Measurement of storage capacity

- It is important that you understand a little about the measurements used to define storage capacities:

Bit: Computer are digital. This means they work by processing ones and zeros. The basic one or zeros is called a bit of information.

Byte: There are eight bits in a Byte.

KB - Kilobyte: There are approximately a 10^3 bytes in a KB

MB - Megabyte: There are approximately a 10^6 bytes in a MB

GB - Gigabyte: There are approximately a 10^9 bytes in a GB

TB - Terabyte: There are approximately a 10^{12} bytes in a TB

Types of storage media

- These days there are a range of storage media to choose from including CDs, DVDs, USB flash drives, memory cards, internal hard disks, external hard disks, network drives and on-line file storage. Some are more suitable than others for a particular job, for instance a flash drive is great for quickly transferring relative small amounts of data from one computer to another.

Internal hard disks

- All PCs are supplied with an internal hard disk. This is where the operating system (such as Windows) is stored. It is also were you store your data. When you install new applications, they are copied from CD or DVD to your

internal hard disk.

External hard disks

- As the name suggests these are secondary hard disks that you can plug into your computer. They are normally connected via a USB cable. They are available in a range of speeds and storage capacities and are an ideal way to backup your data, such as photos or movies.

CDs

- Most computers are now supplied with a CD-ROM (Compact Disc - Read Only Memory) drive. CD-ROM discs look exactly like music CDs but contain computer data instead of music. The

DVDs

- Short for "Digital Versatile Disk". Similar to CD-ROM drives but allows you to use DVD disks, which contain vastly more information than a traditional CD-ROM disk. These also transfer the data from the disk to the computer far faster, allowing you to watch movies on your computer screen. A CD-ROM can store 650 MB of data, while a single-layer, single-sided DVD can store over 4 GB of data.

Recordable CD & DVD drives

- CD-ROMs are read-only devices, but increasingly people are purchasing a special type of CD drive unit which allows you to record data, music or video to your own CDs. These devices require the purchase of special CDs to which you can write, called CD-R (Compact Disc - Recordable).

USB flash drives (memory sticks)

- Flash drives plug into the USB port and when viewed via the Windows

Explorer, look just like any other drive. They are supplied in a range of sizes with the 1 GB devices being a very cheap way of transferring relatively small amounts of data between computers.

Memory cards

- A memory card (also called a flash memory card) is a card containing memory chips that is often used in devices such as digital cameras, telephones, music players, video game consoles, GPS system and similar devices where there is a need to store data in a compact from, often using a battery power source. There are a number of different types of memory cards with different storage capacities.

Input Devices

- Input devices allow you to input information to the computer and include things such as the keyboard and mouse.

Keyboard

- The keyboard allows you to type information into the computer. It has evolved over the years. The keyboard is built into laptop computers but is a separate item if used with a Desktop computer. They can be connected via cables or may be wireless.

Mouse

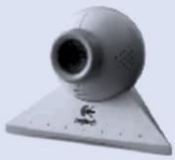
- When using an operating system, such as Microsoft Windows, you use the mouse to select drop down menus, to point and click on items, to select items and to drag and drop items from one place to another.

Scanners

- A scanner allows you to scan printed material and convert it into a file format which may be used within the PC. You can scan pictures and then manipulate these inside the PC using a graphics application of your choice. In addition, you can scan printed text and convert this not just to a picture of the text but also to, actual text which can be manipulated and edited as text within your word-processor. There are a number of specialist programs, generically called OCR (Optical Character Recognition) programs which are specifically designed for converting printed text into editable text within your applications.

Web cams

- Ever since it was invented, the Web has become increasingly interactive. You can now use a small digital movie camera (a Web cam) mounted on the PC monitor to allow two-way communication involving not just text communication but sound and video communication as well. While not yet considered a standard piece of PC kit, it is only a matter of time ...



Digital cameras

- A digital camera can be used in the same way a traditional camera can, but instead of storing images on rolls of film which require developing, the images are stored digitally in memory housed within the camera. These pictures can easily be transferred to your computer and then manipulated within any graphics programs which you have installed on your computer. Currently they are limited by the quality of the image recorded and the number of pictures which you may store within the camera.

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من أمثلة علي input devices

Microphones

- Early voice recognition systems offered very poor results, due to the limitations of the software combined with hardware limitations. It takes an awful lot of CPU processing power to convert the spoken word into text which appears on the screen. Things are changing rapidly however and recent systems allow you to talk to a PC and see text appear on the screen. Most of these systems require an initial training period, where you train the software to respond to your particular voice. Whilst still not perfect this is a key technology of the future.

Output devices

- Includes items such as screens (monitors), printers, speakers and headphones.

Traditional computer monitor

- An output device. The original computer monitors were TV type screens on which you viewed your programs. They were supplied in different sizes, common sizes range from 15" to 21" screens. You should be aware that poor quality or badly maintained monitors could harm your eyesight.

Flat screen computer screens

- Traditional computer monitors are based on the same sort of technology which is used within a television screen.

More recently, flat screen computer screens have become available. These take up a lot less room on a desk and use less energy than the traditional, more bulky monitors.

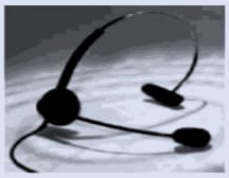
- You should be aware that often if you specify a screen of a certain size, say a 17-inch screen, this is the size measured diagonally, not horizontally across the screen. If you are upgrading you should also ask for the "visible viewing area" of the screen.

Projection devices

- These are projection devices which can be attached to your computer and are useful for displaying presentations to a group of people. They are best used in combination with presentation programs, such as Microsoft PowerPoint. They are used within education and are also very popular for sales presentations. The price of these devices has dropped dramatically recently. When purchasing one of these devices the two things to look out for are the resolution (go for a minimum of XGA) and the brightness of the lamp (the brighter the better). Other factors to be aware of are the quietness (or otherwise) of the device, as well as the cost of replacement bulbs.

Speakers and headphones

- Most computers are sold with the capability to add a pair of speakers to your system unit. In fact, in some cases, the computer screen may have speakers built directly into the unit. This enhances the value of educational and presentation products and can now be considered a standard PC component.
- Most computers allow you to plug headphones into one of the computer sockets. This is useful as the sound will then not annoy others in the room.



Speech synthesizers

- A recent development is the ability not only to display text on a monitor but also to read the text to you. Thus, you could receive a text email from a colleague and the system could read that email to you. This is of enormous benefit to the visually impaired when using a computer. On the flip side, it is now possible to use a microphone to talk to the computer and for the computer to directly convert the spoken word into text which will be displayed within say your word-processor. Whilst these systems are far from foolproof they are getting better as more advanced software is being made available.

Printers

- Most data is printed once you have created it and there are a vast number of different printers available to accomplish this. Most common are ink jet and laser printers both of which can now produce colored output (at a cost).

Laser printers

- Laser printers produce high print quality at high speed. They are called "laser printers" due to the fact that they contain a small laser within them. There is a wide range of laser printer manufacturers and one buzzword to be aware of is Postscript, a type of printer which is designed to give very high quality reproduction of pictures.

Color laser printers

- Originally, most laser printers would only print in black and white (mono). More recently color laser printers have dropped in price and are entering wide spread use. While many of these produce excellent results, you should be aware of the fact that the "price per page", especially if you are using a lot of color on a page can be very high compared to the cost of printing in black and white.

Inkjet printers

- Inkjet printers work by using tiny jets to spray ink onto the paper. Inkjet printers are very quiet in operation and produce print quality comparable to that of laser printers, though laser printers still have the edge in terms of speed. Inkjet printers are ideal for low volume printing where high quality print is required and

speed is not a high priority, e.g. printing letters in a small office or in the home.

Input and output devices

Some devices are both input and output devices.

- A modem can be used for downloading information from web sites and receiving emails. It can also be used for uploading and sending emails. A touch screen can display a menu system (output device), and accept input when people touch the menus displayed on the screen.